Forecasting System Imbalance in the Electricity Grid

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Introduction

The objective of this project was to forecast the sign of electricity grid system imbalance using historical time-series data. The analysis was based on a single time series provided in an Excel file, `grid_data.xlsx`.

The task focused on predicting whether the system imbalance would be positive or negative one and two time steps ahead. Model performance was evaluated using sign prediction accuracy rather than the numerical accuracy of the forecasts.

The analysis involved exploratory time-series analysis, comparison of statistical forecasting methods, model selection, and out-of-sample evaluation. Particular attention was given to temporal dependence, potential seasonal patterns, and the trade-off between predictive performance and computational complexity.

Data Preparation and Exploratory Data Analysis

The dataset contained 8,833 observations and was well-structured, requiring minimal preprocessing. The only necessary transformation was to construct a timestamp by combining the existing day and time columns. The data were then divided chronologically into training and test sets, with the first 90% of observations used for model development and the final 10% reserved for out-of-sample evaluation.

All exploratory analysis, visualization, model selection, and parameter estimation were performed exclusively on the training set. The test set was used only to evaluate the predictive performance of the selected models, with particular emphasis on sign prediction accuracy. This chronological split preserves the temporal ordering of observations and reflects the intended forecasting setting.

The following plots summarize the main characteristics of the training data and provide a basis for selecting appropriate forecasting methods.

Figure 1 shows the overall behavior of the time series. The observations fluctuate around zero, with no immediately apparent long-term trend. The series exhibits considerable short-term variation, broadly resembling a stochastic process, although some oscillatory behavior may indicate underlying temporal structure. These initial observations motivate a more detailed investigation of autocorrelation and potential seasonal patterns.

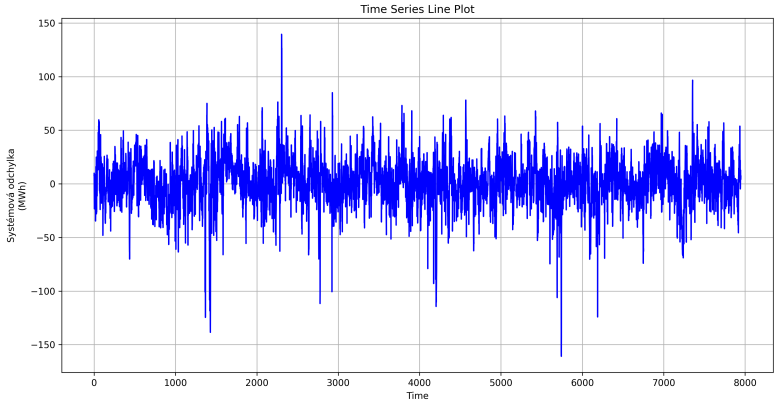


Figure 1: Time series of system imbalance.

To examine short-term dynamics in greater detail, Figure 2 presents the series over a single representative day. Similar plots were inspected for several days, but no consistent pattern was readily apparent from these individual daily trajectories. This suggests that any recurring intraday structure may be difficult to identify from individual days alone.

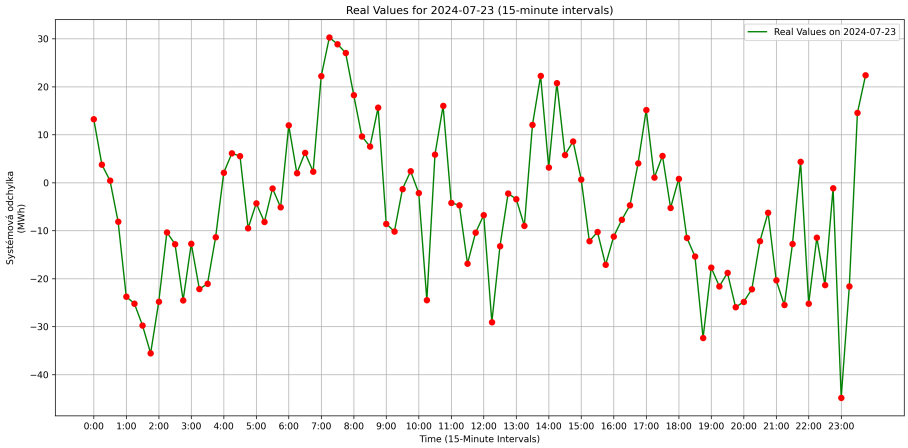


Figure 2: System imbalance over a representative day.

Figure 3 shows the average system imbalance for each 15-minute interval of the day, calculated across the available training observations. The plot also displays the corresponding upper and lower one-standard-deviation bounds around the

interval means. This aggregation helps identify potential intraday patterns that may be obscured by substantial variation at the individual-observation level.

The average profile exhibits some variation across the day, suggesting the possibility of recurring intraday structure. However, the dispersion around the mean profile is considerable, indicating substantial variability between observations and limiting the extent to which a consistent daily pattern can be inferred from the averages alone.

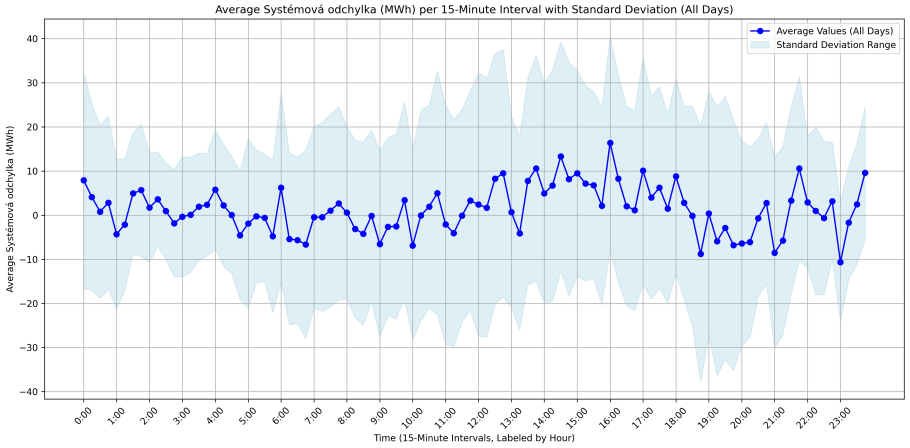


Figure 3: Mean system imbalance by 15-minute interval, with one-standard-deviation bounds.

Figure 4 presents the daily average system imbalance. Examining daily averages helps reveal lower-frequency movements that may be less apparent in the original high-frequency series. This view complements the intraday analysis, although daily aggregation can conceal substantial variation within each day.

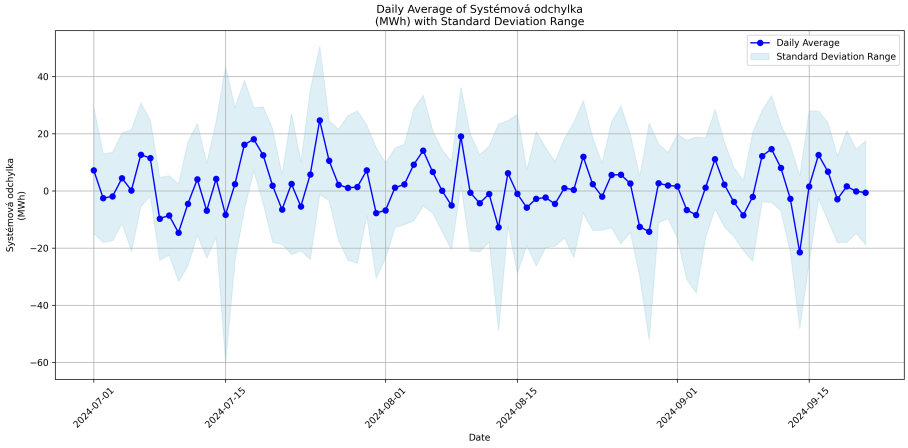


Figure 4: Daily average system imbalance.

To assess temporal dependence, the autocorrelation function (ACF) was examined. Figure 5 indicates statistically significant autocorrelation across multiple lags, extending to approximately lag 19 according to the original analysis. This suggests that consecutive observations are not independent and that past values may contain useful information for short-term forecasting.

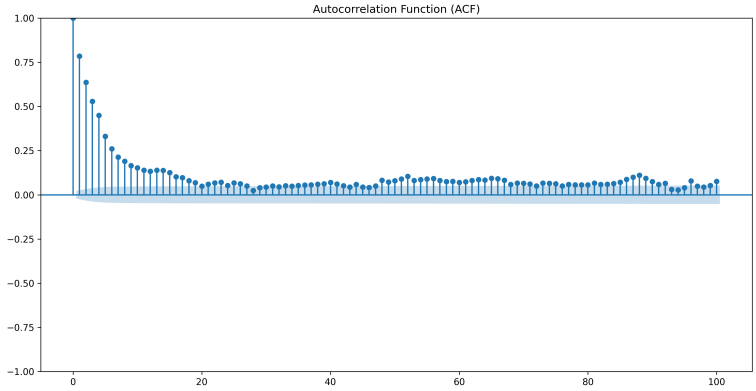


Figure 5: Autocorrelation function of the system imbalance time series.

The partial autocorrelation function (PACF), shown in Figure 6, provides additional insight into the dependence structure after accounting for intermediate lags. The original analysis identified significant partial autocorrelation through

approximately lag 3. Together, the ACF and PACF suggest that autoregressive and moving-average models are worth investigating.

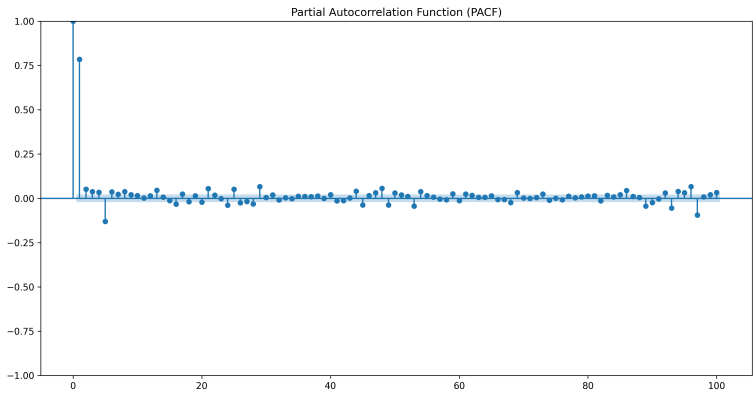


Figure 6: Partial autocorrelation function of the system imbalance time series.

Overall, the exploratory analysis suggests that the series fluctuates around zero, exhibits substantial short-term variability, and contains temporal dependence. Although the plots provide some indication of potential intraday structure, its strength is uncertain relative to the overall variability. These findings motivate the evaluation of autoregressive and seasonal time-series models in the following section.

Model Selection and Forecasting

The exploratory analysis indicated that the time series fluctuates around zero without a pronounced long-term trend, while exhibiting substantial temporal dependence and potential intraday seasonality. Given the forecasting objective and the short prediction horizons, autoregressive and seasonal time-series models were considered promising approaches.

Classical decomposition methods were not pursued further. Although useful for describing trend and seasonal components, their contribution to forecasting in this particular setting appeared limited. The absence of a pronounced trend and the substantial variability remaining after decomposition suggested that a considerable proportion of the series would remain unexplained. Instead, the analysis focused on Box–Jenkins models, which explicitly account for temporal dependence and are well suited to short-term forecasting.

Seasonal ARIMA (SARIMA)

SARIMA models extend the ARIMA framework by incorporating seasonal autoregressive and moving-average components. Given the apparent intraday structure and the possibility of recurring daily patterns, a seasonal period of 96 observations was considered appropriate, corresponding to one day of 15-minute observations.

However, estimating models with this seasonal period presented computational and memory constraints, limiting the range of specifications that could be evaluated successfully. The most complex specification successfully estimated was

$$\text{SARIMA}(0,0,0)(2,0,2)_{96}.$$

This model achieved sign prediction accuracies of 75.31% and 74.86% for one-step-ahead and two-step-ahead forecasts, respectively. Although these results were promising, the model search was constrained by computational limitations. Further exploration of seasonal specifications and parameter combinations could potentially improve predictive performance.

ARMA

ARMA models provide a computationally simpler alternative by capturing temporal dependence without explicitly modeling seasonal components. The ACF and PACF suggested that relatively short autoregressive orders and longer moving-average orders were worth investigating.

A grid search was performed over candidate ARIMA specifications with $p \in \{0,1,2\}$, $d = 0$, and $q \in \{0,\ldots,18\}$. The Akaike Information Criterion (AIC) was used to compare successfully estimated models, and the specification with the lowest AIC was selected. This procedure identified an ARMA(2,16) model.

The model achieved sign prediction accuracies of 75.65% for one-step-ahead forecasts and 74.18% for two-step-ahead forecasts. Its performance was comparable to that of the seasonal model, with slightly higher accuracy at the first forecast horizon but slightly lower accuracy at the second.

Given its simpler specification and competitive predictive performance, ARMA(2,16) was selected as the final model. The selection was based on the AIC within the candidate range considered, rather than an exhaustive search over all possible model specifications. Nevertheless, SARIMA remained a promising alternative, particularly if computational constraints could be addressed and seasonal model selection further optimized.

Potential Extensions

Several alternative approaches could be investigated in future work. Alternative seasonal specifications could be explored to capture recurring intraday patterns

while reducing computational demands. GARCH-type models could be useful if the series exhibits time-varying conditional volatility, while threshold models such as SETAR could capture nonlinear dynamics associated with different imbalance regimes.

If additional explanatory variables became available, multivariate models such as VAR or machine-learning approaches, including neural networks, could also be considered. Another promising direction would be to combine forecasts from multiple models, potentially assigning greater weight to models that perform better on recent observations. Such extensions would require further empirical evaluation and, where relevant, additional knowledge of electricity-grid operation and regulation.

Conclusion

The ARMA(2,16) model was selected as the final model due to its competitive predictive performance and relative simplicity. It achieved sign prediction accuracies of 75.65% and 74.18% for one-step-ahead and two-step-ahead forecasts, respectively. Although these results are promising, there remains considerable scope for improvement.

The substantial variability of the time series suggests that some of its behavior may be difficult to predict using historical values alone. Further improvements could be pursued in several directions:

- Further optimization and evaluation of seasonal ARIMA specifications.
- Investigation of alternative time-series models and nonlinear approaches.
- Incorporation of additional explanatory variables and observations.
- Use of domain knowledge about electricity-grid operation and regulation.
- Combination of forecasts from multiple models to improve predictive performance.

Overall, this project demonstrates the application of time-series analysis, statistical model selection, and out-of-sample evaluation to a practical forecasting problem. Further research incorporating additional data, domain knowledge, and alternative modelling approaches could provide a basis for more accurate and robust predictions.